

Insurance Portfolio Overview (Motor Claims Risk Dashboard)

Total Policies

678K

Overall Claim Frequency

10.1%

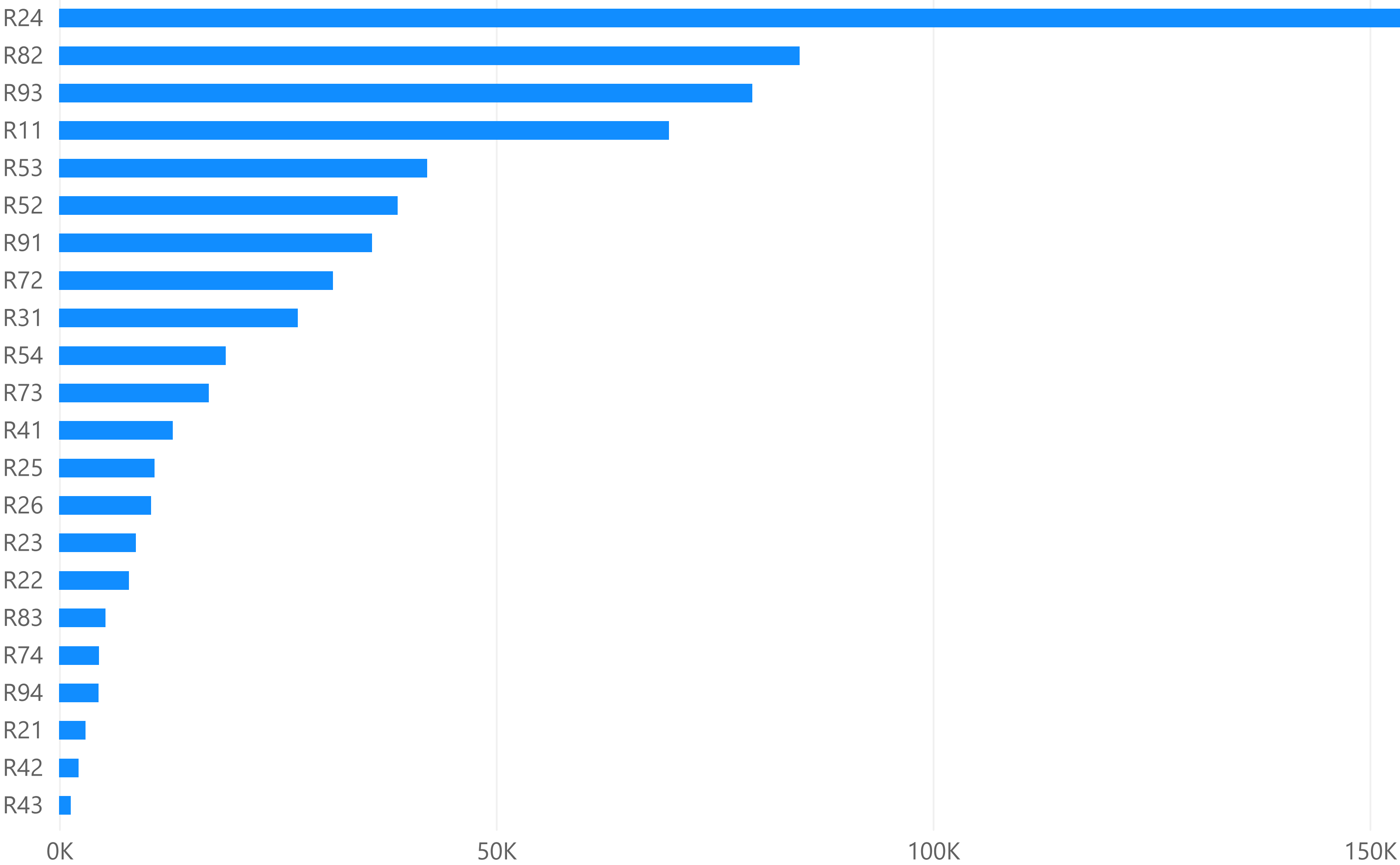
Average Severity

2K €

Portfolio Loss Ratio

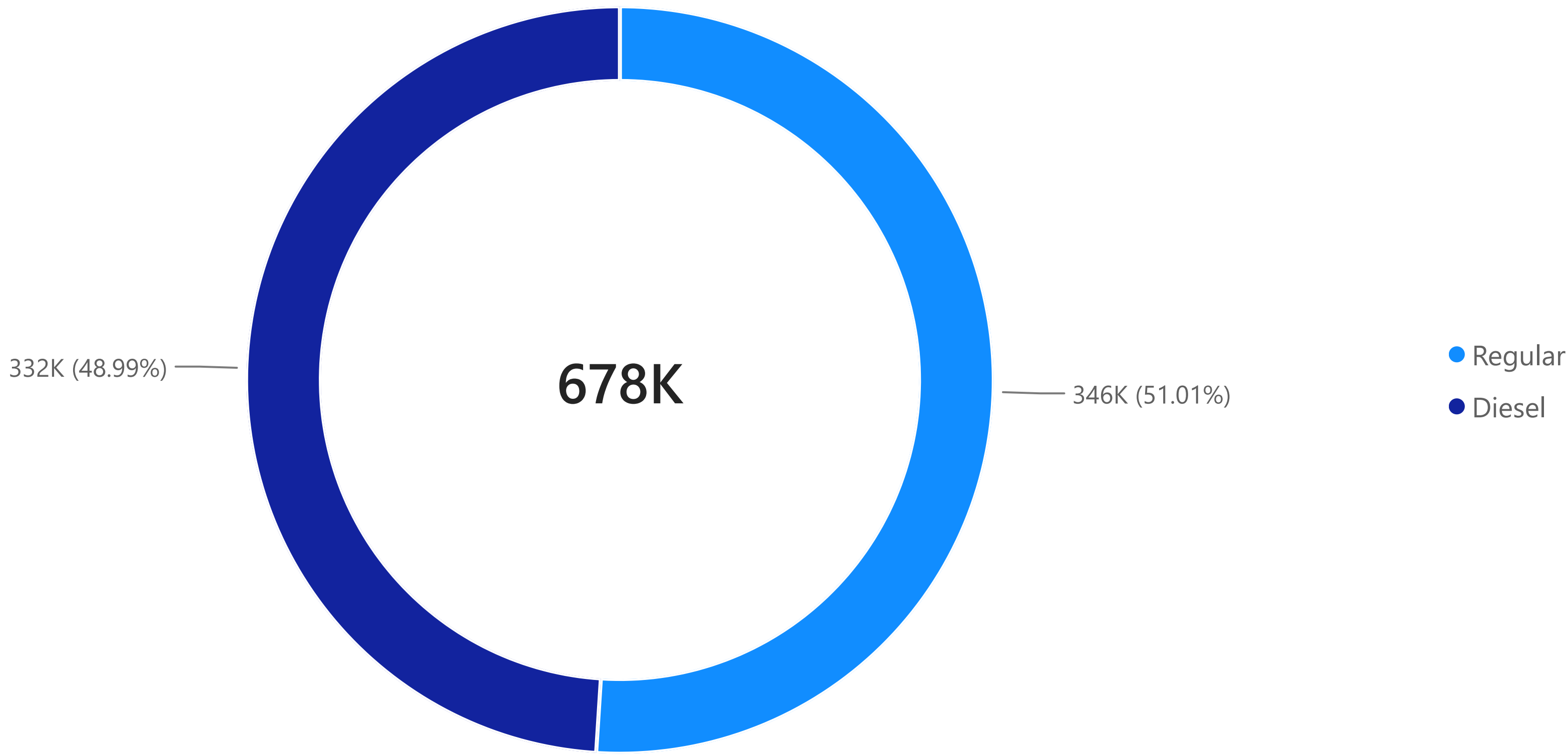
56.9%

Policy Count by Region



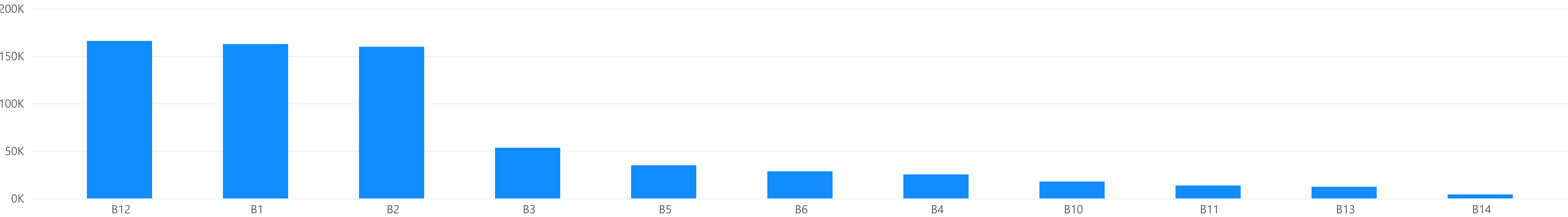
Vehicle Fuel Type Split

Total Policies by VehGas



Policy Count by Vehicle Brand

Total Policies by VehBrand



Claim Frequency by Risk Segment

Area

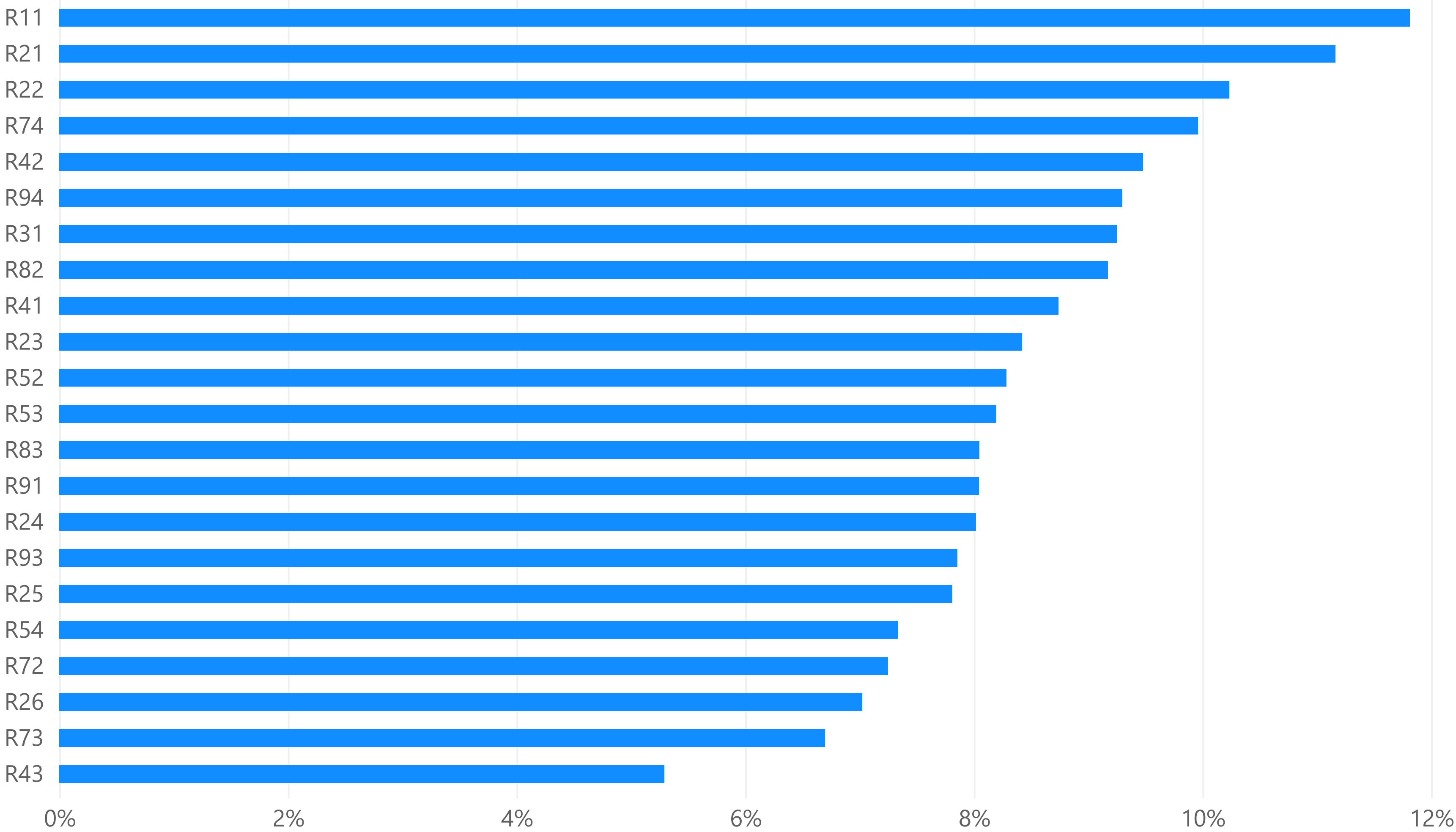
A

VehGas

Diesel

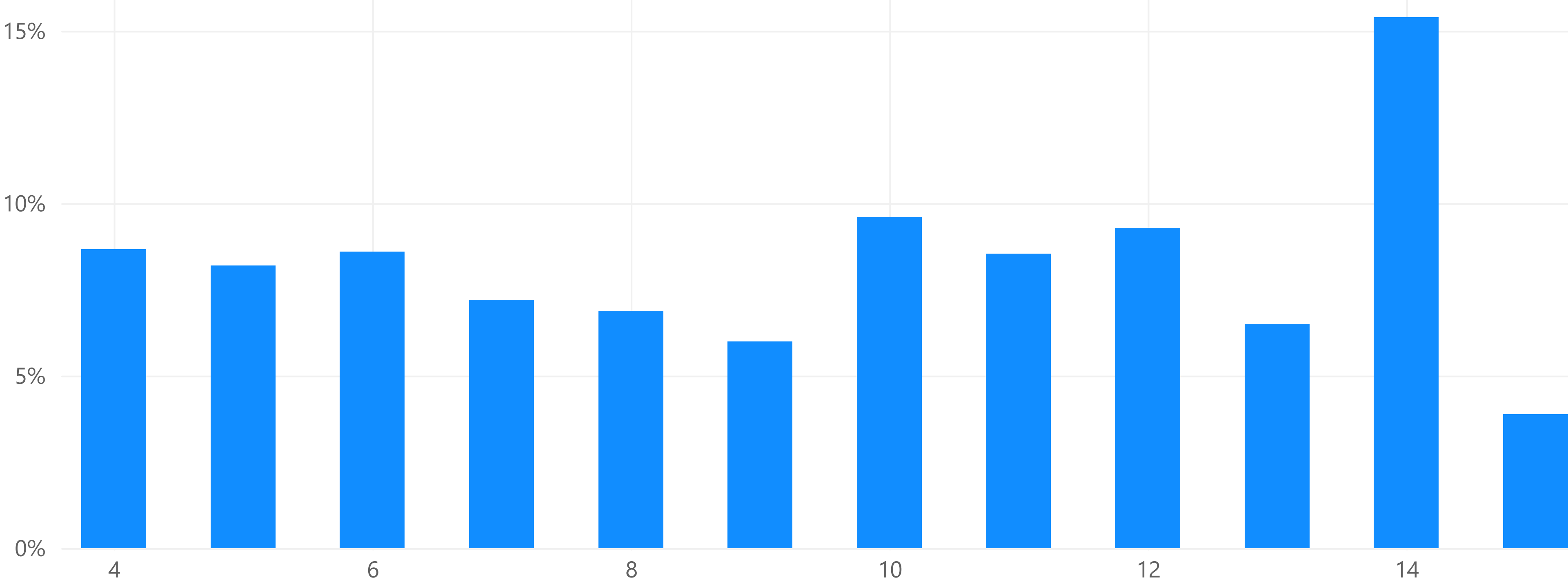
Claim Frequency by Region

Overall Claim Frequency by Region



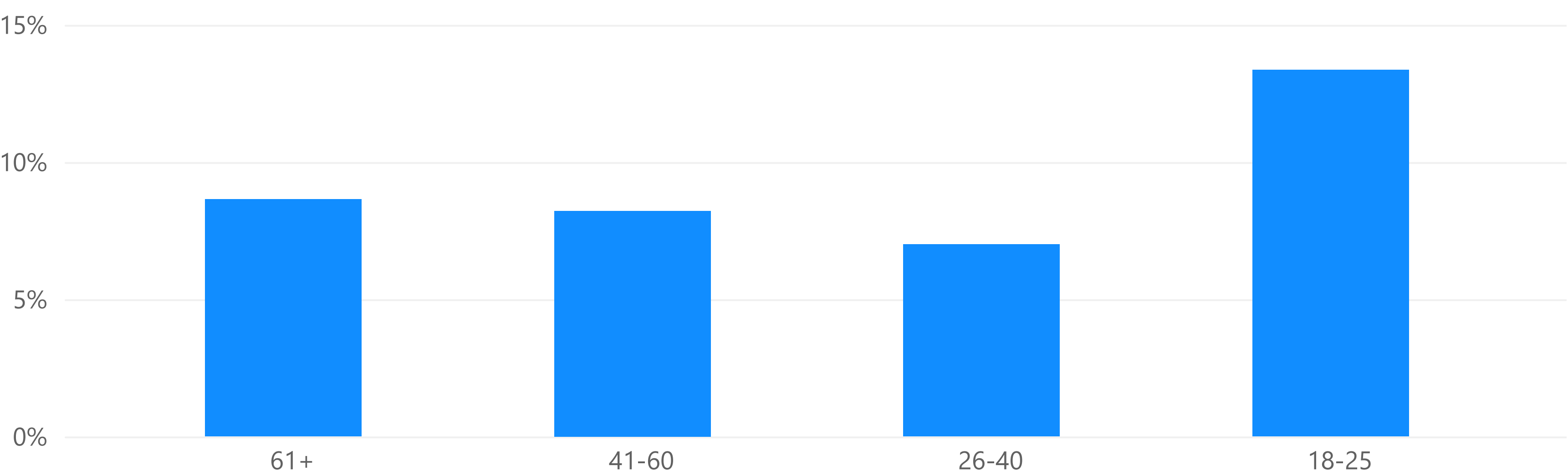
Claim Frequency by Vehicle Power

Overall Claim Frequency by VehPower



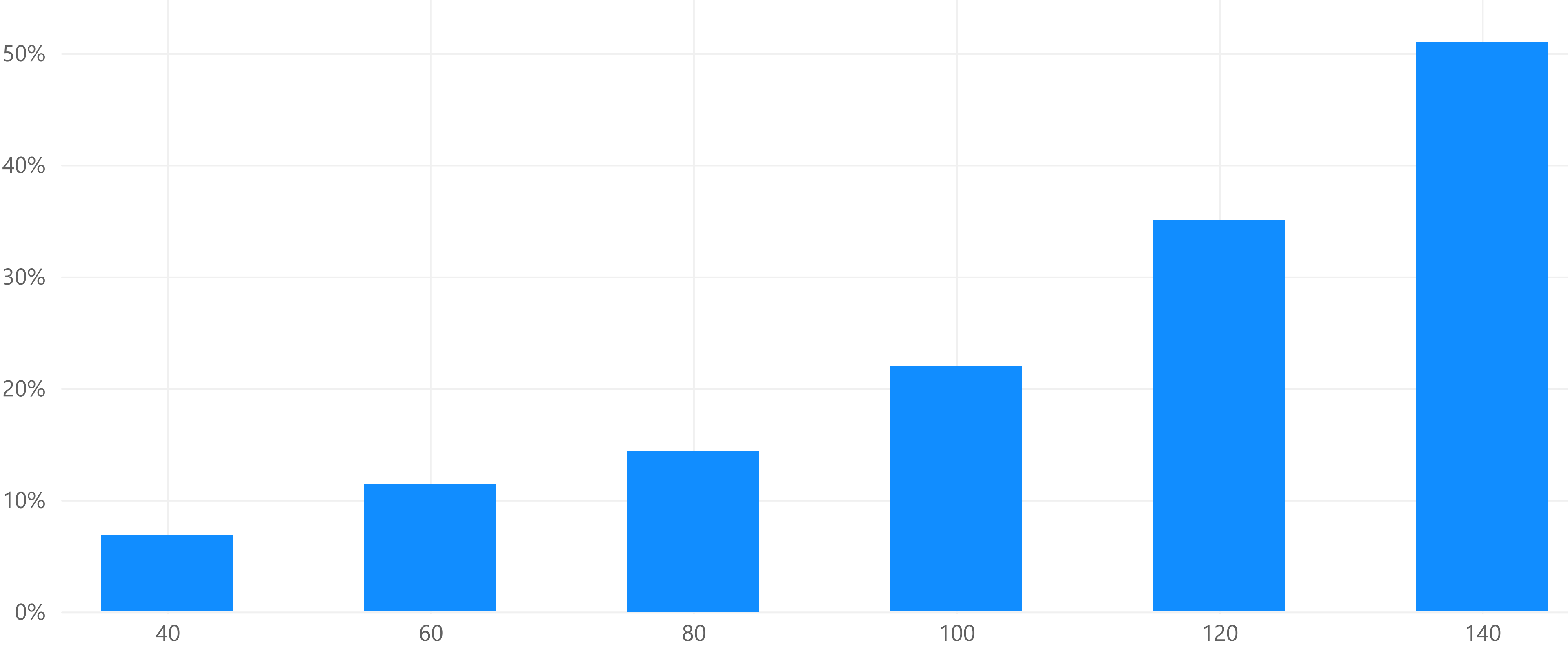
Claim Frequency by Driver Age Band

Overall Claim Frequency by DrivAgeBand



Claim Frequency by BonusMalus Score

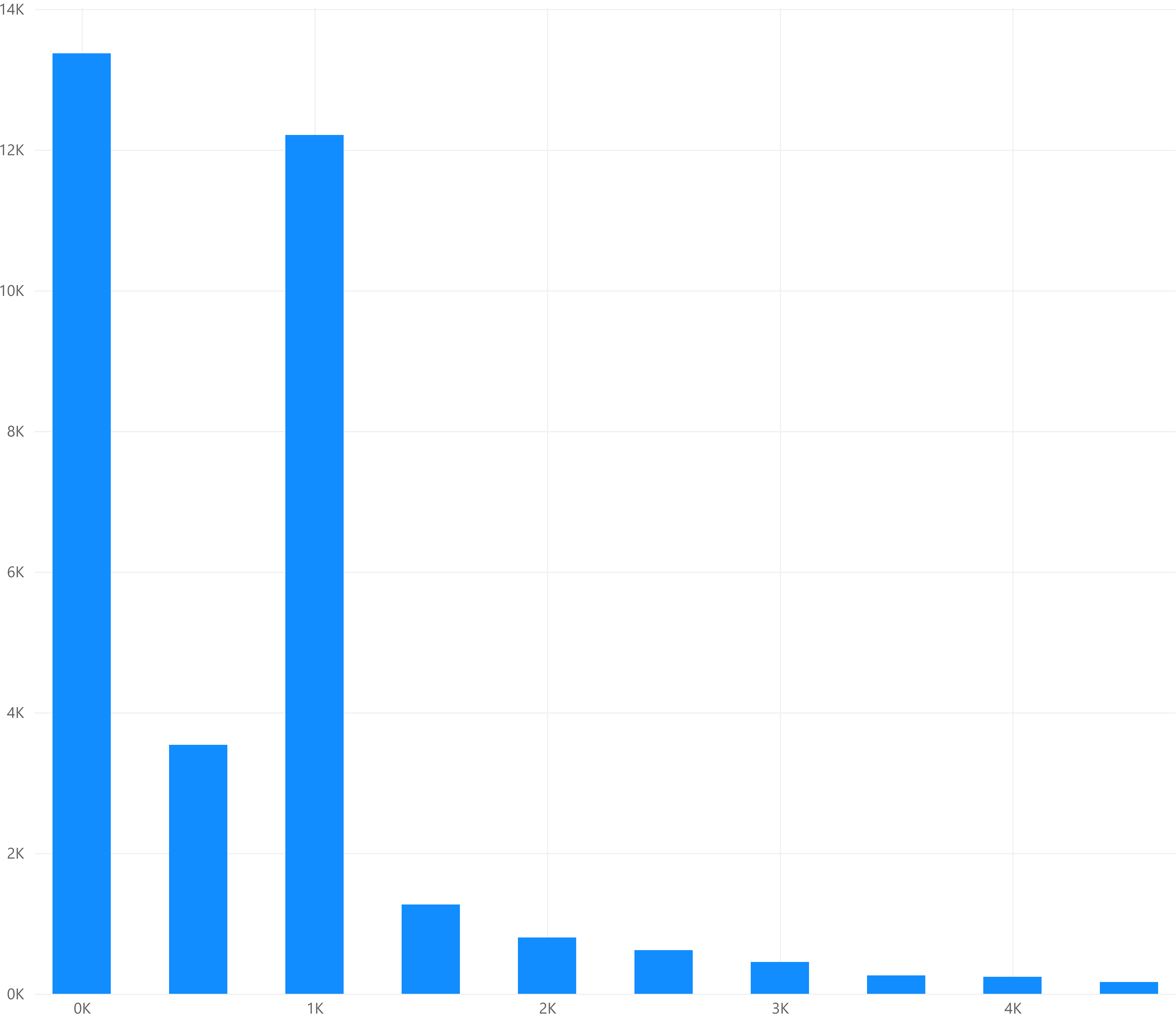
Overall Claim Frequency by BonusMalus (bins)



Claim Severity, Model Performance & Key Drivers

Claim Severity Distribution (Policies With Claims)

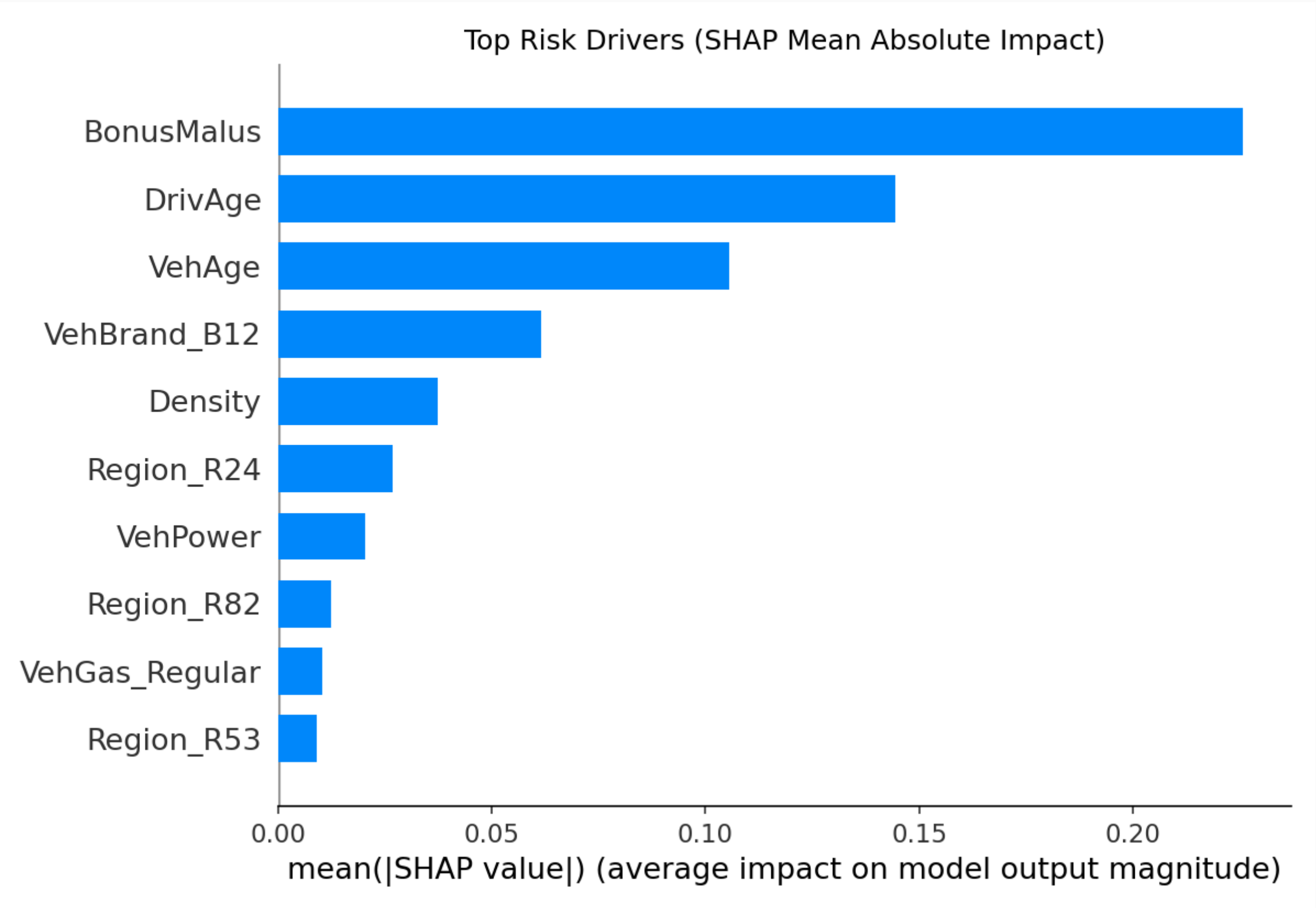
Count of IDpol by AvgSeverity (bins)



Model Performance: Baseline vs. LightGBM

Model	Metric	LightGBM	Baseline
Frequency	Poisson Deviance	0.35	0.36
Severity	Gamma Deviance	1.57	1.50
Severity	Log MAE	0.85	1.02

LightGBM improved frequency prediction by ~3% over baseline Poisson regression (lower deviance = better fit). For severity, LightGBM roughly matched the baseline on deviance but meaningfully reduced typical prediction error (log MAE), suggesting better performance on the bulk of claims even if it doesn't fully capture the largest outlier claims.



BonusMalus is the dominant predictor of claim frequency, followed by VehPower, DrivAge, and Area. Higher BonusMalus, higher vehicle power, younger drivers, and urban areas all push predicted risk upward, consistent with standard motor insurance rating factors.